Apply the Sum and Difference Formulas for Sine and Cosine
Apply the Sum and Difference Formulas for Tangent

Sum and Difference Formulas

$$\sin(u+v)=\sin u\cos v+\cos u\sin v \qquad \cos(u+v)=\cos u\cos v-\sin u\sin v$$
$$\sin(u-v)=\sin u\cos v-\cos u\sin v \qquad \cos(u-v)=\cos u\cos v+\sin u\sin v$$

$$\tan(u+v)=\frac{\tan u+\tan v}{1-\tan u\tan v} \qquad \tan(u-v)=\frac{\tan u-\tan v}{1+\tan u\tan v}$$

Use an addition or subtraction formula to find the exact value.

1. $\sin 15°$

2. $\cos\frac{7\pi}{12}$

3. $\tan 75°$

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4. $\sin\left(-\frac{\pi}{12}\right)$

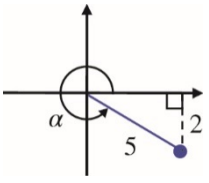
5. $\sin 65° \cos 20° - \cos 65° \sin 20°$

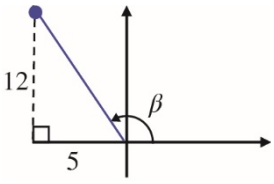
6. $\dfrac{\tan\frac{7\pi}{9} - \tan\frac{\pi}{9}}{1 + \tan\frac{7\pi}{9}\tan\frac{\pi}{9}}$

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7. Find the exact value of $\cos(\alpha - \beta)$.

$\sin\alpha = -\frac{2}{5}$ for α in Quadrant IV and $\tan\beta = -\frac{12}{5}$ for β in Quadrant II.





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Use the Sum and Difference Formulas to Verify Identities

Verify the identity.

8. $\sin\left(x-\frac{3\pi}{2}\right)=\cos x$

9. $\cos(x+y)-\cos(x-y)=-2\sin x\sin y$

10. $\tan\left(x+\frac{\pi}{4}\right)\tan\left(x-\frac{\pi}{4}\right)=-1$

11. $\frac{\sin(x+y)}{\cos x\cos y}=\tan x+\tan y$

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Write a Sum of $A\sin x$ and $B\cos x$ as a Single Term

This formula is called a reduction formula because it "reduces" the sum of two trigonometric functions into one term.

For real numbers A, B, and x,

$$A\sin x + B\cos x = k\sin(x+\alpha)$$

where $k = \sqrt{A^2 + B^2}$, and α satisfies $\cos\alpha = \frac{A}{k}$ and $\sin\alpha = \frac{B}{k}$.

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12. Write $11\sin x + 60\cos x$ in the form $k\sin(x+\alpha)$ for $0 \le \alpha < 2\pi$. Round α to 4 decimal places.

13. Check your answer to Exercise 12 by using the sum formula for sine.

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